Tianyuan Lu, PhD

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University of Wisconsin School of Medicine and Public Health
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Google Scholar Profile: https://scholar.google.ca/citations?user=hBnK0YAAAAAJ&hl=en

Education

McGill University, Montreal, QC, Canada — Sept. 2018 – May. 2022
PhD in Quantitative Life Sciences, Statistical Genetics and Genetic Epidemiology Stream

Fudan University, Shanghai, China — Sept. 2014 – Jun. 2018
BSc in Biological Sciences, Biostatistics and Bioinformatics Stream

Employment

University of Wisconsin-Madison, Madison, WI, USA — Aug. 2024 - Present
Assistant Professor (Tenure-Track)
Department of Population Health Sciences & Department of Biostatistics and Medical Informatics
School of Medicine and Public Health

Five Prime Sciences Inc., Montreal, QC, Canada — Dec. 2023 – Present
Consultant

University of Toronto, Toronto, ON, Canada — Feb. 2023 – Jul. 2024
Schmid AI in Science Postdoctoral Fellow

Five Prime Sciences Inc., Montreal, QC, Canada — Jun. 2022 – Sept. 2023
Senior Research Scientist

Research Support (**Active*)

7. *R35 Maximizing Investigators' Research Award (MIRA) (2,071,525 USD, over five years), NIH/NIGMS, 2026

6. Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship (170,000 CAD, over two years), Schmidt Futures, 2023

5. CANSSI Ontario STAGE HostSeq Project Fellowship (10,000 CAD, over one year), Canadian Statistical Sciences Institute, 2022

4. Vanier Canada Graduate Scholarship (150,000 CAD, over three years), Canadian Institutes of Health Research, 2021

3. FRQS Doctoral Training Fellowship (84,000 CAD, over four years), Fonds de Recherche du Québec Santé, 2020

2. Faculty of Medicine Fellowship (12,000 CAD), Department of Medicine, McGill University, 2020 <Declined due to funding from another source>

1. Faculty of Medicine Fellowship / W. R. Lasha Research Fellowship (12,000 CAD, over one year), Department of Medicine, McGill University, 2019

Other Honors and Awards (**Competitive*)

24. *Early Career Researcher Travel Award (500 GBP), International Mendelian Randomization Conference, University of Bristol, 2024
23. *James V. Neel Award (1,000 USD), International Genetic Epidemiology Society, 2023
22. Eric and Wendy Schmidt Conference Travel Award (travel reimbursement valued at 5,000 CAD), Schmidt Futures, 2023
21. *Prix Relève Étoile Jacques-Genest Publication Award (1,500 CAD), Fonds de Recherche du Québec Santé, 2023
20. *ECTS New Investigator Award (750 EUR), European Calcified Tissue Society, 2023
19. *Annual McGill Biomedical Graduate Conference Outstanding Oral Presentation, Second Place Award (350 CAD), McGill University, 2022
18. *Charles J. Epstein Trainee Awards Semifinalist (750 USD), American Society of Human Genetics, 2021
17. *Young Investigator Travel Award (500 USD), American Society for Bone and Mineral Research, 2021 <Declined due to COVID-19 pandemic>
16. McGill Initiative in Computational Medicine Graduate Award (400 CAD), McGill University, 2021
15. Plan de Réussite Graduate Mobility Award (2,000 CAD), McGill University, 2021
14. *McGill MedStar Publication Award / A. W. K. Akerley Award (500 CAD), Department of Medicine, McGill University, 2020
13. Quantitative Life Sciences Travel Award (500 CAD), McGill University, 2019
12. Australian Institute of Bioengineering and Nanotechnology Research Fellowship (2,880 AUD), University of Queensland, 2018
11. *Excellent Graduate of Class 2018 (Top 5%), Fudan University, 2018
10. Visiting Student Research Fellowship (5,000 USD), King Abdullah University of Science and Technology, 2017
9. Scholarship of Summer Institute in Statistical Genetics (tuition exemption valued at 1,050 USD), Department of Biostatistics, University of Washington, 2017
8. *Fosun Pharma Scholarship, First Prize (Top 2%, 5,000 CNY), School of Life Sciences, Fudan University, 2017
7. *Fudan University Student Exchange Fellowship / Dong Fang Scholarship (29 recipients, 6,000 CNY), Fudan University, 2017
6. *Scholarship for Outstanding Students, First Prize/Arawana Scholarship (Top 5%, 10,000 CNY), Fudan University, 2016
5. *Jing Rong Life Sciences Scholarship, First Prize (Top 5%, 3,000 CNY), School of Life Sciences, Fudan University, 2016
4. *Outstanding Student Scholarship in Faculty of Science (Top 30%, 1,800 CNY), Fudan University, 2016
3. *Chinese College Students Math Competition, Third Prize, Chinese Mathematical Society, 2015
2. *Campbell's Scholarship, First Prize (Top 5%, 5,000 CNY), School of Life Sciences, Fudan University, 2015
1. *National Scholarship (Top 2%, 8,000 CNY), Ministry of Education of the People's Republic of China, 2015

Publication Summary

My research focuses on developing and implementing rigorous statistical genetics and genetic epidemiology methods to improve the prevention, diagnosis, and treatment of complex diseases, and translating research findings into new medical care approaches and therapies.

As First/Co-First Author: 25
As Senior/Co-Senior Author: 10
As Co-Author: 33

Research Articles as First or Senior Author (†*Authors contributed equally* ¶*Corresponding author*)

35. Wenmin Zhang¶, **Tianyuan Lu¶**, Robert Sladek, Josée Dupuis, and Guillaume Lettre¶. Robust fine-mapping in the presence of linkage disequilibrium mismatch. Accepted for publication in ***Nature Genetics***.

34. Shihui Peng, Guillaume Butler-Laporte, Sterling C. Johnson, Corinne D. Engelman, and **Tianyuan Lu¶**. Genetic evidence suggests a protective role of immunoglobulin M in Alzheimer's disease. Accepted for publication in ***Alzheimer's & Dementia***.

33. Chen-Yang Su, Masashi Hasebe, Shihui Peng, Dandan Tan, Bangli Cao, Kevin Liang, Takayoshi Sasako, Vincent Mooser, Wenmin Zhang, Sirui Zhou, Satoshi Yoshiji, **Tianyuan Lu¶**, and Guillaume Butler-Laporte¶. Disentangling osteoarthritis-specific genetic effects from obesity to identify novel therapeutic targets. Accepted for publication in ***Nature Communications***.

32. Jerome J. Choi, Corinne D. Engelman, and **Tianyuan Lu¶**. Multi-omics integration of transcriptomics and metabolomics with machine learning uncovers novel risk factors for Alzheimer's disease. Accepted for publication in ***Journal of Alzheimer's Disease***.

31. Yanzhen Cheng, Guillaume Butler-Laporte, Tomoko Nakanishi, and **Tianyuan Lu¶**. Development of a simple clinical score to prioritize detection of severe alpha-1 antitrypsin deficiency with PiZZ genotype. ***Genetics in Medicine*** 2026.
https://doi.org/10.1016/j.gim.2026.102690

30. Wenmin Zhang¶, Satoshi Yoshiji, Robert Sladek, Josée Dupuis¶, and **Tianyuan Lu¶**. Benchmarking Bayesian colocalization methods in validating Mendelian randomization-based target discoveries from circulating proteins for cardiometabolic diseases. ***Genomics, Proteomics & Bioinformatics*** 2026.
https://doi.org/10.1093/gpbjnl/qzag087

29. **Tianyuan Lu¶**, Lily N. Stalter, Kate V. Lauer, Bret M. Hanlon, Wenmin Zhang, and Luke M. Funk. A multi-ancestry polygenic risk score for body mass index predicts longitudinal weight change. ***Genome Medicine*** 2026.
https://link.springer.com/article/10.1186/s13073-026-01741-8

28. **Tianyuan Lu¶**, David Stein, Wenmin Zhang, Yuval Itan, and Andrew D. Paterson. Development of polyphenotypic scores to prioritize detection of G6PD rs1050828-T carriers in African and African American populations. ***Genetics in Medicine*** 2026.
https://doi.org/10.1016/j.gim.2026.101943

27. Wenmin Zhang¶, Qiongshi Lu, and **Tianyuan Lu¶**. Challenges to case-only analysis for gene-environment interaction detection using polygenic risk scores: model assumptions and biases in large biobanks. ***Genetics*** 2026.
https://doi.org/10.1093/genetics/iyag006

26. Jerome J. Choi, Noah Cohen Kalafut, Tim Gruenloh, Corinne D. Engelman, **Tianyuan Lu¶**, and Daifeng Wang¶. Cooperative multi-view integration with a scalable and interpretable model explainer. ***Nature Machine Intelligence*** 2025.
https://www.nature.com/articles/s42256-025-01111-w

25. **Tianyuan Lu¶**, Wenmin Zhang, Fergus W. Hamilton, Guillaume Butler-Laporte, Nicholas J. Timpson, George Davey Smith, and J. Brent Richards¶. Challenges to the Independence Assumption in Mendelian

Randomization due to Sample Selection and Complex Methods. ***The Journal of Clinical Endocrinology & Metabolism*** 2025.
https://doi.org/10.1210/clinem/dgaf305

24. **Tianyuan Lu**¶ and Andrew D. Paterson¶. Estimating effects of serum vitamin B12 levels on psychiatric disorders and cognitive impairment: a Mendelian randomization study. ***Communications Medicine*** 2025.
https://doi.org/10.1038/s43856-025-01043-x

23. **Tianyuan Lu**¶, Wenmin Zhang, Cassiane Robinson-Cohen, Corinne D. Engelman, Qiongshi Lu, Ian H. de Boer, Lei Sun, and Andrew D. Paterson. Improved characterization of gene-environment interactions for vitamin D through variance quantitative trait loci. ***American Journal of Clinical Nutrition*** 2025.
https://doi.org/10.1016/j.ajcnut.2025.01.021

22. **Tianyuan Lu**¶, Despoina Manousaki, Lei Sun, and Andrew D. Paterson¶. Integrative proteogenomic analyses provide novel interpretations of type 1 diabetes risk loci through circulating proteins. ***Diabetes*** 2025.
https://doi.org/10.2337/db24-0380

21. Nazim Rabouhi, Simon Guindon, Emilia Aisha Coleman, H. J. van Heesbeen, Celia M. T. Greenwood¶, **Tianyuan Lu**¶, and Philippe Campeau¶. Assessing in silico Tools for Accurate Pathogenicity Prediction in CHD Nucleosome Remodelers. ***Journal of Molecular Biology*** 2025.
https://doi.org/10.1016/j.jmb.2025.169413

20. Wenmin Zhang, Chen-Yang Su, Satoshi Yoshiji, and **Tianyuan Lu**¶. MR Corge: Sensitivity analysis of Mendelian randomization based on the core gene hypothesis for polygenic exposures. ***Bioinformatics*** 2024.
https://doi.org/10.1093/bioinformatics/btae666

19. **Tianyuan Lu**¶, Yiheng Chen, Satoshi Yoshiji, Yann Ilboudo, Vincenzo Forgetta, Sirui Zhou, and Celia M. T. Greenwood¶. Circulating metabolite abundances associated with risks of psychiatric disorders: a Mendelian randomization study. ***Biological Psychiatry*** 2024.
https://doi.org/10.1016/j.biopsych.2024.04.016

18. **Tianyuan Lu**¶, Vincenzo Forgetta, Sirui Zhou, J Brent Richards, and Celia M. T. Greenwood¶. Identifying rare genetic determinants for improved polygenic risk prediction of bone mineral density and fracture risk. ***Journal of Bone and Mineral Research*** 2023. **Lady Davis Institute Paper of the Month** (December 2023)
https://asbmr.onlinelibrary.wiley.com/doi/10.1002/jbmr.4920

17. **Tianyuan Lu**¶, Tomoko Nakanishi, Satoshi Yoshiji, Guillaume Butler-Laporte, Celia M. T. Greenwood, and J. Brent Richards¶. Dose-dependent association of alcohol consumption with obesity and type 2 diabetes: observational and Mendelian randomization analyses. ***The Journal of Clinical Endocrinology & Metabolism*** 2023.
https://doi.org/10.1210/clinem/dgad324

16. **Tianyuan Lu**¶, Patrícia Pelufo Silveira, and Celia M. T. Greenwood¶. Development of risk prediction models for depression combining genetic and early life risk factors. ***Frontiers in Neuroscience*** 2023.
https://www.frontiersin.org/articles/10.3389/fnins.2023.1143496/full

15. **Tianyuan Lu**¶, Vincenzo Forgetta, J. Brent Richards, and Celia M. T. Greenwood¶. Genetic determinants of polygenic prediction accuracy within a population. ***Genetics*** 2022.

https://doi.org/10.1093/genetics/iyac158

14. Julian St. Pierre†, Xinyi Zhang†, **Tianyuan Lu**†, Lai Jiang†, Linbo Wang, Sahir Bhatnagar, Celia M. T. Greenwood¶, and CANSSI team on Improving Robust High-Dimensional Causal Inference and Prediction Modelling. Considering strategies for SNP selection in genetic and polygenic risk scores. ***Frontiers in Genetics*** 2022.
https://www.frontiersin.org/articles/10.3389/fgene.2022.900595/full

13. **Tianyuan Lu**, Vincenzo Forgetta, Celia M. T. Greenwood, Sirui Zhou, and J. Brent Richards¶. Circulating proteins influencing psychiatric disease: A Mendelian randomization study. ***Biological Psychiatry*** 2022. **Prix Relève Étoile Jacques-Genest / Best Student Publication Award of Québec** (March 2023)
https://doi.org/10.1016/j.biopsych.2022.08.015

12. **Tianyuan Lu**¶, Vincenzo Forgetta, J. Brent Richards, and Celia M. T. Greenwood¶. Capturing additional genetic risk from family history for improved polygenic risk prediction. ***Communications Biology*** 2022.
https://www.nature.com/articles/s42003-022-03532-4

11. **Tianyuan Lu**, Vincenzo Forgetta, Celia M. T. Greenwood, and J. Brent Richards¶. Identifying causes of fracture beyond bone mineral density: Evidence from human genetics. ***Journal of Bone and Mineral Research*** 2022.
https://asbmr.onlinelibrary.wiley.com/doi/10.1002/jbmr.4632

10. Zhen-Hui Wang†, **Tianyuan Lu**†, Ming-Rui Li, Ning Ding, Li-Zhen Lan, Xiang Gao, Ai-Sheng Xiong, Jian Zhang, and Lin-Feng Li¶. Genetic and epigenetic associated with the divergence of *Aquilegia* species. ***Genes*** 2022.
https://www.mdpi.com/2073-4425/13/5/793

9. **Tianyuan Lu**, Vincenzo Forgetta, J. Brent Richards, and Celia M. T. Greenwood¶. Polygenic risk score as a possible tool for identifying familial monogenic causes of complex diseases. ***Genetics in Medicine*** 2022.
https://www.gimjournal.org/article/S1098-3600(22)00717-1/fulltext

8. Célia Escribe†, **Tianyuan Lu**†, Julyan Keller-Baruch, Vincenzo Forgetta, Bowei Xiao, J. Brent Richards, Sahir Bhatnagar, Karim Oualkacha and Celia M. T. Greenwood¶. Block coordinate descent algorithm improves variable selection and estimation in error-in-variables regression. ***Genetic Epidemiology*** 2021. **Highlight of Issue** (December 2021)
https://onlinelibrary.wiley.com/doi/full/10.1002/gepi.22430

7. **Tianyuan Lu**, Andres Cardenas, Patrice Perron, Marie-France Hivert, Luigi Bouchard, and Celia M. T. Greenwood¶. Detecting cord blood cell type-specific epigenetic associations with gestational diabetes mellitus and early childhood growth. ***Clinical Epigenetics*** 2021.
https://clinicalepigeneticsjournal.biomedcentral.com/articles/10.1186/s13148-021-01114-5

6. **Tianyuan Lu**, Vincenzo Forgetta, Haoyu Wu, John R. B. Perry, Ken K. Ong, Celia M. T. Greenwood, Nicholas J. Timpson, Despoina Manousaki, and J. Brent Richards¶. A polygenic risk score to predict future adult short stature among children. ***The Journal of Clinical Endocrinology & Metabolism*** 2021.
https://doi.org/10.1210/clinem/dgab215

5. **Tianyuan Lu**[†], Vincenzo Forgetta[†], Julyan Keller-Baruch, Maria Nethander, Derrick Bennett, Marie Forest, Sahir Bhatnagar, Robin G. Walters, Kuang Lin, Zhengming Chen, Liming Li, Magnus Karlsson, Dan Mellström, Eric Orwoll, Eugene V. McCloskey, John A. Kanis, William D. Leslie, Robert J. Clarke, Claes Ohlsson, Celia M. T. Greenwood, and J. Brent Richards[¶]. Improved prediction of fracture risk leveraging a genome-wide polygenic risk score. ***Genome Medicine*** 2021.
https://genomemedicine.biomedcentral.com/articles/10.1186/s13073-021-00838-6

4. **Tianyuan Lu** and Jessica C. Mar[¶]. Investigating transcriptome-wide sex dimorphism by multi-level analysis of single-cell RNA sequencing data in ten mouse cell types. ***Biology of Sex Differences*** 2020.
https://link.springer.com/article/10.1186/s13293-020-00335-2

3. **Tianyuan Lu**, Sirui Zhou, Haoyu Wu, Vincenzo Forgetta, Celia M. T. Greenwood, J. Brent Richards[¶]. Individuals with common diseases, but with a low polygenic risk score could be prioritized for rare variant screening. ***Genetics in Medicine*** 2020. **Editor's Featured Article** (March 2021)
https://www.nature.com/articles/s41436-020-01007-7

2. **Tianyuan Lu**, Vincenzo Forgetta, Oriana H. Y. Yu, Lauren Mokry, Madeline Gregory, George Thanassoulis, Celia M. T. Greenwood and J. Brent Richards[¶]. Genetic influences on coronary heart disease act on atherosclerosis in type 2 diabetes. ***Cardiovascular Diabetology*** 2020.
https://link.springer.com/article/10.1186/s12933-020-0988-9

1. **Tianyuan Lu**, Kathleen Oros Klein, Inés Colmegna, Maximilien Lora, Celia M. T. Greenwood and Marie Hudson[¶]. Whole-genome bisulfite sequencing in systemic sclerosis provides novel targets to understand disease pathogenesis. ***BMC Medical Genomics*** 2019.
https://doi.org/10.1186/s12920-019-0602-8

Research Articles as Co-Author ([†]***Authors contributed equally*** [¶]***Corresponding author*****)**

33. Zhenqian Wang[†], Chen-Yang Su[†], Jiawen Lu, Masashi Hasebe, Alejandro Mejia-Garcia, Hsuan Megan Tsao, Jingyi Hu, **Tianyuan Lu**, and Sirui Zhou[¶]. Integrative plasma proteomic network analysisidentifies physical activity-associated protein modulespotentially mediating type 2 diabetes risk. ***Cardiovascular Diabetology*** 2026.
https://link.springer.com/article/10.1186/s12933-026-03363-1

32. Basile Jumentier, HuiQi Qu, **Tianyuan Lu**, Kai Liu, Erica Kleinbrink, Kathleen Klein, Wiame Belbellaj, Isabel Gamache, Lauric Ferrat, Guillaume Butler-Laporte, Yangxi Li, Hakon Hakonarson, Wei Wu, Constantin Polychronakos, Celia M. T. Greenwood, and Despoina Manousaki[¶]. Development and validation of a Trans-Ancestry polygenic risk score for Type 1 Diabetes. ***Diabetologia*** 2026.
https://link.springer.com/article/10.1007/s00125-026-06706-5

31. Yuanqin Yang[†], Jiawen Lu[†], Zhenqian Wang[†,¶], **Tianyuan Lu**, Jin Ding, Ruofan Wang, Sirui Zhou, Zhiguang Zhou, and Jingji Hu[¶]. ***Cardiovascular Diabetology*** 2026.
https://link.springer.com/article/10.1186/s12933-026-03203-2

30. Guillaume Butler-Laporte[¶], **Tianyuan Lu**, Sam Morris, Wenmin Zhang, Gavin Band, Fergus W. Hamilton, Amanda Chong, Kuang Lin, Ruth Nanjala, J. Brent Richards, Mei-Hsuan Lee, Ling Yang, Pang Yao, Liming Li, Zhengming Chen, Yang Luo, Iona Y. Milwood, Robin G. Walters, and Alexander J. Mentzer. An accurate genetic colocalization method for the HLA locus. ***HLA*** 2026.
https://onlinelibrary.wiley.com/doi/10.1111/tan.70759

29. Thomas M. Zheng, Yann Ilboudo, **Tianyuan Lu**, Guillaume Butler-Laporte, Tomoko Nakanishi, David Morrison, Darin Adra, Lena Cuddeback, and J. Brent Richards¶. Differential circulating proteomic responses associated with ancestry during severe COVID-19 infection. ***Journal of Proteome Research*** 2025.
https://pubs.acs.org/doi/abs/10.1021/acs.jproteome.4c00956

28. Fergus W. Hamilton¶, David A. Hughes, **Tianyuan Lu**, Zoltán Kutalik, Apostolos Gkatzionis, Kate Tilling, Fernando Pires Hartwig, and George Davey Smith. Non-linear Mendelian randomization: evaluation of effect modification in the residual and doubly-ranked methods. ***European Journal of Epidemiology*** 2025.
https://link.springer.com/article/10.1007/s10654-025-01208-x

27. Amanda M. Marks, Guillaume Butler-Laporte, Satoshi Yoshiji, **Tianyuan Lu**, Dave R. Morrison, Tomoko Nakanishi, Yiheng Chen, Vincenzo Forgetta, Yossi Farjoun, Robert Frithiof, Miklós Lipcsey, Hugo Zeberg, J. Brent Richards, and Michael Hultström¶. Aquaporin 3 genotype modulates the risk of death conferred by plasma osmolality in COVID-19. ***Physiological Genomics*** 2025.
https://journals.physiology.org/doi/full/10.1152/physiolgenomics.00174.2024

26. Satoshi Yoshiji¶, **Tianyuan Lu**, Guillaume Butler-Laporte, Julia Carrasco-Zanini-Sanchez, Yiheng Chen, Kevin Liang, Julian D. S. Willett, Chen-Yang Su, Shidong Wang, Darin Adra, Yann Ilboudo, Takayoshi Sasako, Satoshi Koyama, Tetsushi Nakao, Vincenzo Forgetta, Yossi Farjoun, Hugo Zeberg, Sirui Zhou, Michael Hulström, Mitchell Machiela, Rama Kaalia, Hesam Dashti, Melina Claussnitzer, Jason Flanick, Nicholas J. Wareham, Vincent Mooser, Nicholas J. Timpson, Claudia Langenberg, and J. Brent Richards¶. Integrative proteogenomic analysis identifies COL6A3-derived endotrophin as a mediator of the effect of obesity on coronary artery disease. ***Nature Genetics*** 2025.
https://www.nature.com/articles/s41588-024-02052-7

25. Jiacheng Miao, Yixuan Wu, Zhongxuan Sun, Xinran Miao, **Tianyuan Lu**, Jiwei Zhao, and Qiongshi Lu¶. Valid inference for machine learning-assisted GWAS. ***Nature Genetics*** 2024.
https://www.nature.com/articles/s41588-024-01934-0

24. Ariel Madrigal, **Tianyuan Lu**, Larisa M. Soto, and Hamed S. Najafabadi¶. A unified model for interpretable latent embedding of multi-sample, multi-condition single-cell data. ***Nature Communications*** 2024.
https://www.nature.com/articles/s41467-024-50963-0

23. Yann Ilboudo, Satoshi Yoshiji, **Tianyuan Lu**, Guillaume Butler-Laporte, Sirui Zhou, and J. Brent Richards¶. Vitamin D, Cognition, and Alzheimer's Disease: Observational and Two-Sample Mendelian Randomization Studies. ***Journal of Alzheimer's Disease*** 2024.
http://dx.doi.org/10.3233/JAD-221223

22. Thomas R. Austin†, Maria Nethander†, Howard A. Fink†, Anna E. Törnqvist, Diana I. Jalal, Petra Buzkova, Joshua I. Barzilay, Laura Carbone, Maiken E. Gabrielsen, Louise Grahnemo, **Tianyuan Lu**, Kristian Hveem, Christian Jonasson, Jorge R. Kizer, Arnulf Langhammer, Kenneth J. Mukamal, Robert E. Gerszten, Bruce M. Psaty, John A. Robbins, Yan V. Sun, Anne Heidi Skogholt, John A. Kanis, Helena Johansson, Bjørn Olav Åsvold, Rodrigo J. Valderrabano, Jie Zheng, J. Brent Richards, Eivind Coward, and Claes Ohlsson¶. A protein-based risk score improves hip fracture prediction. ***Nature Aging*** 2024.
https://doi.org/10.1038/s43587-024-00639-7

21. Wenmin Zhang¶ ,**Tianyuan Lu**, Robert Sladek, Yue Li, Hamed S. Najafabadi, and Josée Dupuis¶. SharePro: an accurate and efficient genetic colocalization method accounting for multiple causal signals. ***Bioinformatics*** 2024.
https://doi.org/10.1093/bioinformatics/btae295

20. Thomas R. Austin, Howard A. Fink, Diana I. Jalal, Anna E. Törnqvist, Petra Buzkova, Joshua I. Barzilay, **Tianyuan Lu**, Laura Carbone, Maiken E. Gabrielsen, Louise Grahnemo, Kristian Hveem, Christian Jonasson, Jorge R. Kizer, Arnulf Langhammer, Kenneth J. Mukamal, Robert E. Gerszten, Maria Nethander, Bruce M. Psaty, John A. Robbins, Yan V. Sun, Anne Heidi Skogholt, Bjørn Olav Åsvold, Rodrigo J. Valderrabano, Jie Zheng, J. Brent Richards, Eivind Coward, and Claes Ohlsson¶. Large-scale Circulating Proteome Association Study (CPAS) meta-analysis identifies circulating proteins and pathways predicting incident hip fractures. ***Journal of Bone and Mineral Research*** 2024.
https://doi.org/10.1093/jbmr/zjad011

19. Guillaume Butler-Laporte¶, Joseph Farjoun, Tomoko Nakanishi, **Tianyuan Lu**, Erik Abner, Yiheng Chen, Michael Hulström, Andres Metspalu, Lili Milani, Reedik Mägi, Mari Nelis, Georgi Hudjashov, Estonian Biobank Research Team, Satoshi Yoshiji, Yann Ilboudo, Kevin Y. H. Liang, Chen-Yang Su, Julian D. S. Willett, Tõnu Esko, Sirui Zhou, Vincenzo Forgetta, Daniel Taliun, and J. Brent Richards. HLA allele-calling using whole-exome sequencing identifies 129 novel associations in 11 autoimmune diseases: a multi-ancestry analysis in the UK Biobank. ***Communications Biology*** 2023.
https://www.nature.com/articles/s42003-023-05496-5

18. Maria Nethander†, Sofia Movérare-Skrtic†, Anders Kämpe, Eivind Coward, Ene Reimann, Louise Grahnemo, Éva Borbély, Zsuzsanna Helyes, Thomas Funck-Brentano, Martine Cohen-Solal, Juha Tuukkanen, Antti Koskela, Jianyao Wu, Lei Li, **Tianyuan Lu**, Maiken E. Gabrielsen, Estonian Biobank Research Team, Reedik Mägi, Mari Hoff, Ulf H. Lerner, Petra Henning, Henrik Ullum, Christian Erikstrup, Søren Brunak, DBDS Genomic Consortium, Arnulf Langhammer, Tiinamaija Tuomi, Asmundur Oddsson, Kari Stefansson, Ulrika Pettersson-Kymmer, Sisse Rye Ostrowski, Ole Birger Vesterager Pedersen, Unnur Styrkarsdottir, Outi Mäkitie, Kristian Hveem, J. Brent Richards, and Claes Ohlsson¶. An atlas of genetic determinants of forearm fracture. ***Nature Genetics*** 2023.
https://www.nature.com/articles/s41588-023-01527-3

17. Delnaz Roshandel, **Tianyuan Lu**, Andrew D. Paterson¶, and Satya Dash¶. Beyond apples and pears: Sex-specific genetics of body fat percentage. ***Frontiers in Endocrinology*** 2023.
https://www.frontiersin.org/articles/10.3389/fendo.2023.1274791/full

16. Julian Daniel Sunday Willett, **Tianyuan Lu**, Tomoko Nakanishi, Satoshi Yoshiji, Guillaume Butler-Laporte, Vincent Mooser, Sirui Zhou, Yossi Farjoun, and J. Brent Richards¶. Colocalization of expression transcripts with COVID-19 outcomes is dependent on cell type and cell state. ***Human Genetics*** 2023.
https://link.springer.com/article/10.1007/s00439-023-02590-w

15. Jeffrey C. Y. Yu, Yixiao Zeng, Kaiqiong Zhao, **Tianyuan Lu**, Kathleen Oros Klein, Inés Colmegna, Maximilien Lora, Sahir Bhatnagar, Andrew Leask, Celia M. T. Greenwood, and Marie Hudson¶. Novel insights into systemic sclerosis using a sensitive computational method to analyze whole-genome bisulfite sequencing data. ***Clinical Epigenetics*** 2023.
https://clinicalepigeneticsjournal.biomedcentral.com/articles/10.1186/s13148-023-01513-w

14. Lei Sun¶, Zhong Wang, **Tianyuan Lu**, Teri A. Manolio, and Andrew D. Paterson¶. eXclusionarY: Ten years later, where are the sex chromosomes in GWAS? ***American Journal of Human Genetics*** 2023.
https://doi.org/10.1016/j.ajhg.2023.04.009

13. Kevin Y. H. Liang, Yossi Farjoun, Vincenzo Forgetta, Yiheng Chen, Satoshi Yoshiji, **Tianyuan Lu**, and J. Brent Richards¶. Predicting ExWAS findings from GWAS data: A shorter path to causal genes. ***Human Genetics*** 2023.
https://link.springer.com/article/10.1007/s00439-023-02548-y

12. Satoshi Yoshiji, Guillaume Butler-Laporte, **Tianyuan Lu**, Julian Daniel Sunday Willett, Chen-Yang Su, Tomoko Nakanishi, David Morrison, Yiheng Chen, Kevin Y. H. Liang, Michael Hultström, Yann Ilboudo, Zaman Afrasiabi, Shanshan Lan, Naomi Duggan, Chantal DeLuca, Mitra Vaezi, Chris Tselios, Xiaoqing Xue, Meriem Bouab, Fangyi Shi, Laetitia Laurent, Hans Markus Münter, Marc Afilalo, Jonathan Afilalo, Vincent Mooser, Nicholas J. Timpson, Hugo Zeberg, Sirui Zhou, Vincenzo Forgetta, Yossi Farjoun, and J. Brent Richards¶. Proteome-wide Mendelian randomization implicates nephronectin as an actionable mediator of the effect of obesity on COVID-19 severity. ***Nature Metabolism*** 2023.
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Mentoring

- Graduate students
  Jerome J. Choi (Aug. 2024 – May 2025; co-mentored with Corinne D. Engelman and Daifeng Wang) – PhD in Epidemiology; Graduated in May 2025; Now Senior Scientist at AbbVie
  Jovana Jovanovska (Aug. 2024 – Jul. 2025; co-mentored with Corinne D. Engelman) – MSc in Epidemiology; Graduated in Jul. 2025; Now Scientist at Epic Systems
  Ou Ren (Sept. 2025 – Present) – PhD student in Epidemiology
  Shihui Peng (Sept. 2025 – Present) – PhD student in Epidemiology

- Postdoctoral researchers
  Yanzhen Cheng, MD (Aug. 2025 – Aug. 2026) – Visiting Scholar; Now Associate Chief Physician at Zhujiang Hospital
  Chen-Yang Su, PhD (Apr. 2026 – Present) – Postdoctoral Fellow

- Undergraduate students
  Xinyuan Hong (Sept. 2024 – May 2025) – Now undergraduate student at Cornell University
  Yifei Wu (Feb. 2025 – May 2026) – Now PhD student at McGill University
  Stella Guan (Sept. 2025 – Present)

Teaching

Department of Population Health Sciences, University of Wisconsin-Madison 2025, 2026
POPHLTH 551 Introduction to Biostatistics for Population Health
Principal Instructor

Department of Population Health Sciences, University of Wisconsin-Madison 2024, 2025, 2026
POPHLTH 805 Advanced Epidemiology: Causal Inference in Epidemiological Studies
Guest Lecturer

Department of Biostatistics and Medical Informatics, University of Wisconsin-Madison 2025, 2026
BMI 800 Becoming a Biomedical Data Scientist
Guest Lecturer

International Genetic Epidemiology Society Annual Conference Workshop 2022
Epidemiology and architecture of large genetic biobanks: hands-on illustrations
Workshop Lecturer

Academic Services

Journal of Alzheimer's Disease 2026 - Present
Editorial board member

BMC Medicine 2024 - Present
Editorial board member

Population Health Sciences PhD and MS Admission Committee 2024 - Present
Committee member

American Society of Human Genetics Annual Conference Nov. 2024
Platform session co-chair "Long-read sequencing offering new insights into neurological disease"

American Society of Human Genetics Annual Conference Jun. 2024
Conference abstract reviewer "Complex Traits and Polygenic Disorders II: Diabetes, obesity, metabolic syndromes, diseases of internal organs and of the endocrine system"

American Society for Bone and Mineral Research Conference Jun. 2024
Conference abstract reviewer "Omics Approaches for Musculoskeletal Biology, Diseases and Treatments"

American Society of Human Genetics Annual Conference Nov. 2023
Platform session co-chair "Novel tools and data resources for genetic data analysis"

American Society of Human Genetics Annual Conference Oct. 2022
Platform session co-chair "Genetic impacts on the epigenome and beyond"

International Genetic Epidemiology Society Annual Conference Sept. 2022
Platform session co-chair "Biobanks and polygenic risk scores"

International Genetic Epidemiology Society Annual Conference Oct. 2021
Poster session co-chair